

Solving Linear Systems by Substitution

Substituting a given isolated variable, isolating first, special cases, and word models

Directions: Solve each system **by substitution**. Show the substituted equation, the value you solve for first, and the back-substitution. Write each answer as an ordered pair (x, y) — or, for a word problem, as a labelled sentence. Check every answer in the equation you did *not* substitute into.

Worked pattern: From $y = x + 1$ and $2x + y = 7$: replace y in the second equation to get $2x + (x + 1) = 7$, so $3x + 1 = 7$, $x = 2$. Back-substitute: $y = 2 + 1 = 3$. Answer $(2, 3)$; check in $2x + y$: $4 + 3 = 7$ ✓

1. $y = 2x$ and $3x + y = 15$

Answer: _____

2. $y = x - 4$ and $2x + 3y = 3$

Answer: _____

3. $2x + y = 11$ and $3x + 2y = 18$

(Neither equation is solved for a variable yet — isolate one first.)

Answer: _____

4. $x = 3y + 1$ and $2x - 5y = 7$

Answer: _____

5. $x + y = 7$ and $3x - y = 5$

Answer: _____

6. A school play sold 12 tickets to one family. Adult tickets cost 8 dollars each and child tickets cost 5 dollars each, and the family paid 81 dollars in all. Let x be the number of adult tickets and y the number of child tickets. Write the system, then solve it by substitution.

Answer: _____

7. $y = 3x - 2$ and $2x + y = 5$

(The solution is not a pair of whole numbers — keep it in fraction form.)

Answer: _____

8. $x - 3y = -1$ and $2x + y = 12$

Answer: _____

9. $y = 2x + 3$ and $4x - 2y = 1$

(Substitute as usual. If the variables disappear, say what the resulting statement tells you about the system, and describe the graph.)

Answer: _____

10. $y = 3x - 4$ and $6x - 2y = 8$

(If the variables disappear here too, decide what *this* statement means, then give one specific pair that solves the system.)

Answer: _____

11. The perimeter of a rectangle is 46 m. Its length is 5 m more than twice its width. Let x be the width and y the length, both in metres. Write the system, then solve it by substitution to find both dimensions.

Answer: _____

12. A chemist mixes a 20% acid solution with a 50% acid solution to make 12 L of a 30% acid solution. Let x be the litres of 20% solution and y the litres of 50% solution. The acid alone must balance: $0.2x + 0.5y = 3.6$. Write the second equation, then solve the system by substitution.

Answer: _____